

PREM TIWARI

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PROFESSIONAL SUMMARY

Backend and data engineer in Bengaluru with **3 years 8 months** of experience. I started on real-time wind-farm IoT at Vestas, including OPC-UA and MQTT pipelines built from scratch to move on-prem SCADA data to the cloud over a secure, encrypted path, and now build data platforms at Smartsheet across US, EU, and AU. I ship projects end to end: API development, scaling Docker services, Databricks and Snowflake on AWS, OpenSearch for logging, and Datadog for alarms. I stay with the work after it ships, from on-call and cost through to how other teams get access.

EDUCATION

B.Tech, Information Technology — Jabalpur Engineering College, Jabalpur, M.P. Jul 2019 – Apr 2023

PROFESSIONAL EXPERIENCE

Software Engineer II — *Smartsheet*

Oct 2025 – Present

- Built config-driven ETL data pipelines for different sources on Databricks with Python and Spark Structured Streaming across US, EU, and AU: **1,045 jobs / 2,756 tasks** over 11 sources (MySQL CDC, DynamoDB, Kinesis, Salesforce API, NetSuite, Stripe), plus a Snowflake to Delta Lake backfill, shipped with Databricks Asset Bundles (CI/CD).
- Migrated multi-task Apache Spark streaming jobs on Databricks from all-purpose to job clusters across US, EU, and AU production, cutting compute costs by **\$300K+ annually**.
- Landed Change Data Capture (CDC) from 82 sharded Aurora PostgreSQL databases (Aurora PostgreSQL → Flink → Kinesis → Delta tables): **876 jobs / 1,547 tasks**, 282 streaming, feeding a vector database for AI and LLMs.
- Built a high-availability observability log pipeline (CloudWatch → subscription filter → Kinesis → Logstash → OpenSearch) ingesting **45–50 TB/day** across 50–100 Kinesis streams and 1,000+ shards, with sub-second search.
- Built audit automation for Databricks, Snowflake, Jenkins, and AWS, plus onboarding docs and an access-request dashboard so other teams self-serve access — cutting weeks of manual effort to days or hours. Drove on-call tickets from **~40 to single digits** and daily ingestion failures by **~34%**.

Data Engineer — *Vestas*

Feb 2024 – Sep 2025

- Designed a real-time alarm platform for **~150–200 farms / 19k turbines** on Kafka, Redis, and Kinesis (Docker) — Protobuf-encoded alarms from Kafka; built the REST API serving turbine status from DynamoDB with **1–2s lag**, plus a cursor-paginated file-download API (DocumentDB + S3); streamed alarms via Kinesis into Delta Lake for analytics.
- Built OPC-UA and MQTT pipelines from scratch after an AWS Greengrass POC fell short, pushing on-prem SCADA data to the cloud over a secure, encrypted path — 22 Kafka partitions ingesting **~10k data points/sec**, with local-server buffering; enhanced data availability by **96%**.
- Re-built a legacy Databricks streaming job as a Dockerized Kubernetes service (MSK → Kinesis) with KEDA autoscaling on Kafka lag, cutting Delta Lake ingestion latency from minutes to **10–20 seconds**.
- Optimized Databricks and PySpark ETL pipelines for solar and wind farm data, cutting processing time and latency by **~62%** by switching the source from S3 to Kinesis.

Assistant Data Engineer — *Vestas*

Feb 2023 – Jan 2024

- Built a monitoring system surfacing data latency across source services and pinpointing bottlenecks; reduced API latency by **36%**.
- Automated the previously manual, accuracy-based ML model registration to AWS S3, saving the team **12 hrs/week**; raised unit-test coverage to **95%**.
- Delivered Databricks LTS upgrades across **50+ data pipeline jobs** with data-quality testing and zero pipeline failures.

TECHNICAL SKILLS

Languages: Python, Go, C++, SQL

Backend: REST APIs, distributed systems, Docker, Kubernetes

Data Engineering: Apache Spark, PySpark, Spark Structured Streaming, Apache Kafka, Apache Flink, Databricks, Delta Lake, Snowflake, Airflow, dbt, ETL, ELT, data pipelines, data warehousing, data modeling, data lakes, Change Data Capture (CDC), OPC-UA, MQTT, SCADA

AWS & Cloud: Amazon Web Services (AWS), EC2, S3, Lambda, Kinesis, Amazon MSK, SNS, SQS, Glue, API Gateway, VPC, CloudWatch, OpenSearch, Logstash

Databases: Aurora PostgreSQL, Aurora MySQL, PostgreSQL, MongoDB, DynamoDB, Redis, vector database

DevOps & Tools: Docker, Kubernetes, KEDA, Terraform, Terragrunt, Ansible, Jenkins, GitHub Actions, GitLab CI, Git, CI/CD, Grafana, Datadog, MLflow

ACHIEVEMENTS

- **CodeChef:** 5-star, Global Rank 8, College Rank 1 (Starters 70)
- **LeetCode & GeeksforGeeks:** 450+ problems solved
- **TCS CodeVita:** Round 2 Global Rank 83
- **GFG Job-A-Thon 6:** Global Rank 76

RECOGNITION

- Public kudos at Smartsheet from Engineering Manager for ownership; recognized by CTO, Chief AI Officer, SVP, and senior ICs after leading stretch work equivalent to an **extra quarter** — including delivering in **one week** a project a contract team had been working on for two months.
- Won an internal Smartsheet hackathon: prototyped usage analytics that surface which product capabilities a customer needs from real usage — pitched as a retention and expansion play, reusable across similar accounts.

CERTIFICATIONS

- **Snowflake:** Collaboration & Marketplace, Data Application, Data Lake, Data Warehousing, Data Engineering